

Term paper 2

BAN403

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# 1 Problem Definition and System Description

## 1.1 Process architecture and operational analysis

The Miller Pain Treatment Center (MPTC) operates as an outpatient chronic pain clinic within the Eastern Hospital Outpatient Center. The primary process is the delivery of pain treatment to outpatient visitors, and the main flow unit is the patient moving through a sequence of clinical and administrative activities (Laguna & Marklund, 2019). The secondary flow unit is the patient's medical record, which precedes and enables each clinical interaction. The main process can be divided into three sub-processes. The first one being patient intake, then clinical assessment and finally patient discharge. The systems entry boundary is triggered when the signs in at the front desk, and the exit boundary is crossed when the patient completes the check-out process and leaves the facility. The center handles three types of patients: New (30%) require extensive clinical interaction and arrive 30 minutes early, Return (40%) follow the same full path with shorter activity times and Follow-up (30%) bypass resident-driven steps entirely, routing directly to the PA. Process flow is shown in Table A.1 in the appendix.

The AMC context adds significant structural complexity by introducing four resident-driven activities shown in figure 1 of the article (Chambers & Williams, 2017, Figure 1). These steps reflect the AMC's dual mandate as a care provider and medical education center, resulting in longer, resource-intensive process where the Attending is involved multiple sequential and parallel activities. From the customer perspective, patients primarily require minimal waiting times and high-quality clinical outcomes. However, these patients must be balanced against AMC's perspective, being that these requirements must coexist with the clinic's educational mandate, which extends patient cycle times to facilitate clinical training.

### 1.1.1 Design approach

Dr. Weems approach to managing the MPTC is best characterized as incremental improvement, rather than a full redesign of the process (Laguna & Marklund, 2019). Instead of restructuring the fundamental architecture of the AMC, he suggests targeted measures within the existing process. He suggests pre-processing to be able to reduce Resident Review and Teach times, also suggest schedule optimization to better match appointments intervals to service time variability, also an early PSC start to eliminate front-desk delays for the first patient and finally a late-arrival policy adapted from his private practice experience. Each of these ideas change a specific step or parameter but without altering the overall process or the roles of staff that is involved. This is

consistent with the incremental improvement paradigm described in chapter 1, where changes are made at the margin to improve performance within an existing design rather than questioning the design itself (Laguna & Marklund, 2019).

### 1.1.2 Design enablers

Following Chapter 3 (Laguna & Marklund, 2019, sec. 3.4), four design enablers shape the MPTC. Firstly, people are the key constraint. The Attending is a critical bottleneck shared across three mutually exclusive activities. While residents add capacity, they introduce variability due to competing duties. Conversely, the PA provides a parallel channel that decouples Follow-up patients from the main flow. Secondly information flow is manual and paper-based, which creates strict sequential dependencies. Dr. Weems' pre-processing proposal directly targets this by shifting information review offline. Third, technology plays a limited role, the absence of electronic health records or automated tools restricts information transfer speed and prevents real-time schedule adjustments. Finally, facilities present a structural constraint. With one of the four exam rooms dedicated to the PA, the remaining three rooms create a bottleneck during peak loads, as patients require simultaneous availability of a room and clinical staff.

### 1.1.3 Capacity and bottleneck analysis

Applying the capacity analysis framework from chapter 5, the Attending is the clear bottleneck in the AMC system. To assess this, consider the demand placed on the attending during a four-hour session. From the schedule template shown in table 6 in the case (Chambers & Williams, 2017, Table 6), each session contains 5 New, 7 Return and 5 Follow-up appointments, resulting in 17 patients. For new patients, the attending is required for both Teach, mean 7.85 minutes, and Resident and Attending, mean 12.45 minutes, giving an expected attending time of approximately 20.3 minutes per new patient. For return patients the equivalent figures are 5.17 and 9.24 minutes, resulting in 14.4 minutes. Lastly for follow-up patients, the attending is only required in approximately 50% of the cases for a brief consultation of around 3 minutes, giving an expected contribution of 1.5 minutes per follow-up patient. The total expected attending demand per session is therefore approximately:

$$(5 \cdot 20.3) + (7 \cdot 14.4) + (5 \cdot 1.5) = 209.8 \text{ minutes}$$

This will result in above the 240 minutes available in a four-hour session when registration, transition times and overruns are considered. This calculation confirms a mismatch in capacity load at the Attending level, this even before accounting for the high variability in

activity times.

The deterministic 15-minute scheduling interval severely aggravates this capacity mismatch. While new patients require an expected process time of roughly 50 minutes, the highly variable Attending step (standard deviation = 12.21 min) mathematically guarantees that early overrun cascade, inflating waiting times for all subsequent patients. This illustrates the relationship between high utilization, variability and waiting time described in chapter 5 (Laguna & Marklund, 2019, p. 168-169). Additionally, resident availability acts as a secondary capacity constraint, where any absences increases individual workloads, directly delaying the critical Teach and Resident & Attending steps that feed primary bottleneck.

## **1.2 Opportunities for improvement**

### **1.2.1 Process-oriented analysis of weakness**

Applying the process-oriented perspective from chapter 2 (Laguna & Marklund, 2019), three structural weaknesses can be identified. First, the AMC flow contains extensive sequential dependencies involving the Attending, creating a compounding chain of waiting. Resident Review occurs while the patient already occupies an exam room, forcing preparation time into the patients cycle time as pure non-value-adding delay. Second, the 15-minute scheduling interval is mismatched with the actual service time variability, we see that the Attending step alone has a standard deviation exceeding 12 minutes, meaning any early overrun propagates forward through the entire session. Third, the 10% no-show rate result in irrecoverable idle capacity, since cancellations are discovered too late to fill the slot, reducing throughput without reducing staff workload.

### **1.2.2 Improvement opportunities using design principles**

The design principles outlined in section 3.6 Laguna and Marklund, 2019, p. 100-101 provide a structured basis for evaluating Dr. Weems proposed interventions and identifying additional opportunities. These workflow design principles directly motivate the pre-processing proposal. By assigning each patient to a resident the evening before the visit, the resident will be able to complete the case review offline, eliminating the Resident Review step during the clinic session. Also, the Attending can conduct some of the case-specific teaching offline, reducing the Teach step during the session. As Dr. Weems estimates pre-processing will cut the average time for both activities in half, if this happens it would reduce the unit load on the Attending, which is the primary bottleneck identified in section 3.1a.

### **1.2.3 Schedule work based on its critical characteristics and eliminating buffers**

This principle supports the case of schedule optimization. Scheduling all appointment types at 15-minute intervals fails to reflect the high variability in New patients service times. A redesigned template allocating longer slots to New patients and shorter slots to Follow-up patients would better match actual service time distributions. This is well suited for simulation evaluation without disrupting real clinic operations. Similarly, the principle of eliminating non-value-adding buffers motivates the early Patient Service Coordinator (PSC) proposal. Having a PSC available at 7.30 a.m. ensures the first patient enters the clinical flow on time, which is critical since any initial delay propagates forward through the entire session.

### **1.2.4 Design the process for the dominant flow, not for the expectations**

The late-arrival policy successfully implemented in the private practice, where late arrivals was cut in half within 12 months, address patient-driven schedule disruption by protecting the dominant flow from expectations (Laguna & Marklund, 2019, p. 99). Transferring this policy to the AMC introduces uncertainty due to the different patient demographic and institutional culture, but its impact on schedule adherence can be evaluated by varying the arrival time distribution in the simulation.

### **1.2.5 Examine process interactions to avoid suboptimization**

A further opportunity not explicitly proposed by Dr. Weems is reducing teaching time variability across attending physicians. Since Teach occupies the Attending, and through the Attending all downstream patients, high variability in this step has an outsized impact on system performance, a classic case of suboptimization when viewed in isolation (Laguna & Marklund, 2019, p. 99-100). Standardizing the teaching interaction or moving part of it offline through pre-processing could reduce this variability and improve schedule predictability.

## **2 Conceptual Model and Simulation Design**

### **2.1 Derivation from system analysis**

The conceptual model is directly derived from the process analysis made in section 1. The system boundary is defined as the AMC clinic session, spanning from patient arrival to exit after checkout. The private practice is not explicitly modeled but serves as a reference baseline in the qualitative analysis. The flow unit is the individual patient, who carries a type attribute assigned at the generation and used to determine routing throughout the

model.

## 2.2 Process representation

The process flow is shown in Figure B.1 in the appendix, developed as a flowchart based on the process design tools in chapter 4 (Laguna & Marklund, 2019, p. 118-120). All patients enter through a common intake path before being routed to one of the three paths based on patient type. New and Return patients follow the full AMC clinical path. This is Resident Review, Resident and Patient, Teach and Resident and Attending before proceeding to check-out. Follow-up patients are routed to the PA, after which a probabilistic branch determines whether an in-person consultation is required before check-out.

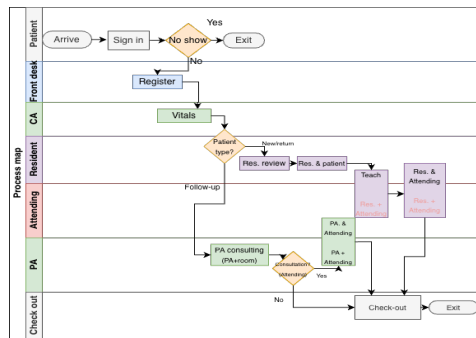


Figure 2.1: Process map

A decision point for no-show patients is modelled after Sign In, with a 10% probability of exit. This reflects the case description that 10% of scheduled appointments are cancelled, and that the clinic learns of these cancellations too late to fill the slot, meaning the appointment time is lost but no service is used. Arrivals are not modelled as a stochastic process, instead patients are generated according to the fixed schedule template shown in Table 6 of the case (Chambers & Williams, 2017, Table 6).

## 2.3 Modelling assumptions

Several simplifying assumptions are made to make the model manageable. First, Sign in is modelled as instantaneous, as it represents a time stamp event rather than a service activity consuming staff time. Second, the three residents are modelled as a shared resource pool with capacity 3 rather than as individually tracked agents. The patient entity seizes a resident from the pool at Resident Review and holds the resource through the entire sequence until Resident and Attending is complete. This reflects the case description that each resident follows a single patient through the full interaction. Third, teaching time variability across attending physicians is not modelled explicitly, a single fitted distribution is used for the Teach step representing an average across physicians. Fourth, the PA

examination room is modeled as a separate resource from the three rooms used for New and Return patients, and as consistent with the case we note that the PA's room is not available for other types of visits.

## 2.4 Mapping to JaamSim

The mapping from conceptual components to JaamSim implementation is summarized in Table 2 found in the appendix. Each activity is implemented as Server object. Routing decisions are implemented as Branch objects. A Queue object precedes each Server, with first in first out priority discipline. Routing decisions are implemented as Branch objects, and deterministic appointment arrivals are implemented using an EventSchedule object linked to an EntityGenerator. Patients who do not show up exit through an EntitySink after the no-show branch, while patients completing Check-out exit through the same sink via a Statistic object that records total time in system.

Four resources are implemented as Resource objects. Resource acquisition and release are handled through Seize and release objects. For new and return patients, a resident and an exam room are seized before Resident Review and held through Resident and Attending is complete. The Attending is seized before Teach and released only after Resident and Attending, ensuring it cannot be simultaneously engaged in more than one activity, directly reflecting the bottleneck identified in section 3.1a. For Follow-up patients, the PA exam room is seized before the PA step, and the Attending is seized for only 50% of cases requiring a consultation.

## 3 Input analysis

### 3.1 Inputs

The stochastic inputs given to the model are divided into two categories: service times and routing decisions. Service times are all administrative and clinical processes done, while routing decisions capture the different paths patients go through the system. A third category, patient arrivals, are simplified to be deterministic, as they follow the patient schedule from Table 6 in the case (Chambers & Williams, 2017). All identified stochastic inputs are listed in Table D.1.

### 3.2 Extracting relevant data

Service time data for the model is gathered from the spreadsheet given in the case (Chambers & Williams, 2017), which holds 90 - 100 observations of the variables used. Discrete distributions used for routing decisions such as no-show rate, patient type and



follow-up consultation are taken from Dr. Weems observations in the case description. Both teaching variables and resident review (new) contain observations on 0-minute service times. As consultations always take time, these observations were removed before fitting.

### 3.3 Fitting appropriate statistical distributions

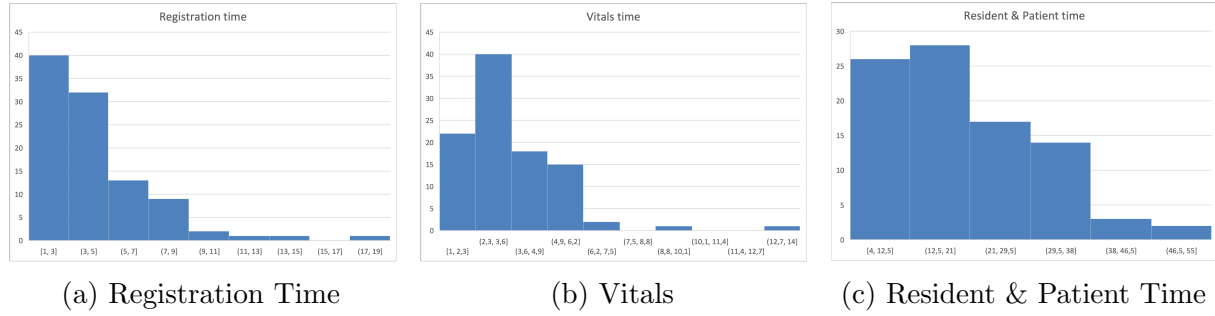


Figure 3.1: Service times

Visual inspection of the histograms can be used to identify the underlying shape of the data, which can then be compared to known theoretical distributions (Laguna & Marklund, 2019, p. 362). The histograms show a right-skewed form with a long tail towards higher values. Figure 3.1 presents histograms for three representative variables. This suggests that distributions supporting right-skewed, positive data such as exponential, gamma, lognormal and weibull should be tested.

All model parameters were fitted using maximum likelihood estimation (MLE), with a fixed location of 0. This is done to avoid negative distributions, as service times in minutes can not hold negative values.

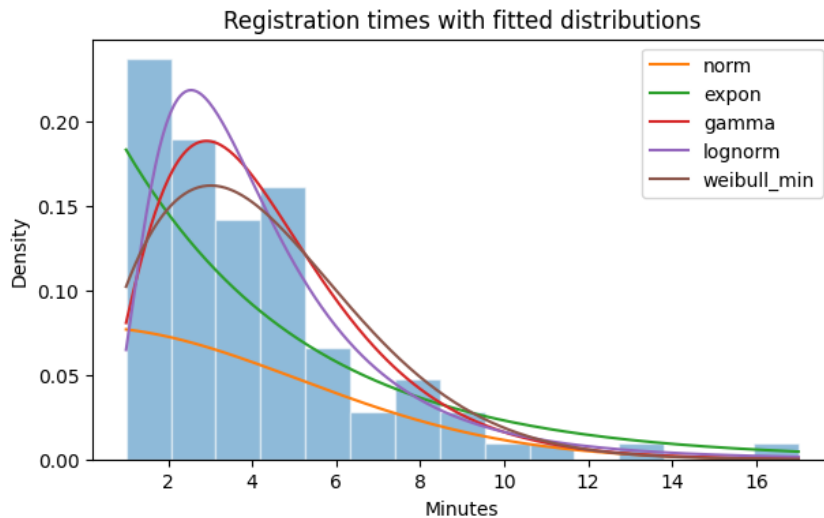


Figure 3.2: Visual fit of distributions

Figure 3.2 shows the histogram of registration times with fitted probability density functions. Gamma and lognormal provide the closest visual fit. Visuals and calculations are available in Calculations.ipynb, submitted with the report.

### 3.4 Goodness-of-fit test

The Kolmogorov-Smirnov test measures the largest absolute deviation between the empirical and theoretical cumulative probability distribution functions (Laguna & Marklund, 2019, p. 423). We evaluate each candidate distributions for every variable, selecting the one with the lowest KS-statistic among the ones passing the 5% significance level.

Gamma is selected for the variable registration, as it has the lowest KS-statistic with a sufficient significance level. As no candidate distribution passes for vitals, we pick the distribution yielding the lowest statistic, which is weibull. Table 3.1 provides results for all variables.

### 3.5 Discussing input assumptions

Multiple assumptions are made regarding the input data. First, time is treated as continuously, although the spreadsheet presents them as integer minutes. Second, zero-valued observations are assumed to be a discrepancy rather than actual service times of zero minutes. Third, to avoid negative and zero-valued service times (as time in minutes can only be positive), all distributions are fitted with a location parameter of 0. Finally, service times are assumed to be independent for all patients and static throughout the simulation period.

### 3.6 Final input model

| Variable                      | Distribution | KS     | p-value |
|-------------------------------|--------------|--------|---------|
| Registration                  | Gamma        | 0.1129 | 0.1481  |
| Vitals                        | Weibull*     | 0.2028 | 0.0005  |
| Physicians Assistant          | Lognormal    | 0.0661 | 0.7602  |
| Check Out                     | Gamma        | 0.0949 | 0.3202  |
| Resident Review (New)         | Lognormal    | 0.1000 | 0.3084  |
| Resident & Patient (New)      | Lognormal    | 0.0852 | 0.5040  |
| Teach (New)                   | Lognormal    | 0.0855 | 0.5281  |
| Resident & Attending (New)    | Gamma        | 0.0739 | 0.6759  |
| Resident Review (Return)      | Lognormal    | 0.0747 | 0.6046  |
| Resident & Patient (Return)   | Weibull      | 0.0677 | 0.7235  |
| Teach (Return)                | Weibull      | 0.0995 | 0.2791  |
| Resident & Attending (Return) | Lognormal    | 0.0946 | 0.3120  |

\* No distribution passed  $p > 0.05$ , Weibull chosen as best fit

Table 3.1: Final inputs with distributions and parameters

Table 3.1 provides all variables with their distributions and parameters.

## 4 Output Analysis

### 4.1 Design

Along with the AS-IS model, three TO-BE scenarios are evaluated as described in section 1.2. As each session lasts approximately 4 hours, we set the simulation time to 6 hours to ensure that every patient could finish their visit. All models were run with 30 replications, which gives a 95% confidence interval of all replications below 8 minutes. The random seeds used in the probability distributions were consistent for all models, enabling comparison and reducing variance across simulations.

### 4.2 Performance measures

The primary output measure is mean cycle time per patient (Chambers & Williams, 2017), defined as time in minutes from arrival to checkout. This directly affects patient care, and is Dr. Weems' primary focus in optimizing the system. Total patients processed is a secondary measure, to ensure that enhancements made do not affect overall system performance.

### 4.3 Confidence intervals

Table 4.1 presents mean cycle time and 95% confidence intervals for all models across 30 replications. The results show a clear improvement of the system for all modifications, with the combined solution giving the best performance. The reduction in confidence interval show not only an improvement in performance, but also a reduction in variability, suggesting a more consistent and reliable process flow.

| Model                   | Mean   | Std    | CI lower | CI upper |
|-------------------------|--------|--------|----------|----------|
| AS-IS                   | 62.011 | 13.239 | 57.068   | 66.955   |
| TO-BE 1: Scheduling     | 55.297 | 11.079 | 51.160   | 59.434   |
| TO-BE 2: Pre-processing | 56.540 | 12.835 | 51.748   | 61.333   |
| TO-BE 3: Combined       | 49.902 | 9.705  | 46.278   | 53.526   |

Table 4.1: Mean cycle time and 95% confidence intervals across models

### 4.4 Statistical comparison

Table 4.1 shows that none of the TO-BE scenario has overlapping confidence intervals with the AS-IS simulation, suggesting statistically significant differences (Laguna & Marklund, 2019, p. 390).

This suggestion is confirmed by the paired t-test applied to compare the outputs from each simulation, measuring the average deviation between models (Laguna & Marklund, 2019, p. 426-429). Since random seeds are shared for all models, performance difference  $D_i = \bar{x}_i^{TO-BE} - \bar{x}_i^{AS-IS}$  is a result of enhancements made rather than variance. The null hypothesis  $H_0 : \mu_D = 0$  is tested at the 5% significance level with  $n - 1 = 29$  degrees of freedom.

The null-hypothesis is rejected for all scenarios, as the t-statistic is well below the critical value of  $t_{0.025,29} = -2.045$ .

| Comparison               | $\bar{D}$ | CI lower | CI upper | t-statistic | p-value |
|--------------------------|-----------|----------|----------|-------------|---------|
| Scheduling vs. AS-IS     | -6.71     | -8.12    | -5.31    | -9.80       | < 0.001 |
| Pre-processing vs. AS-IS | -5.47     | -6.76    | -4.19    | -8.71       | < 0.001 |
| Combined vs. AS-IS       | -12.11    | -14.41   | -9.81    | -10.76      | < 0.001 |

Table 4.2: Paired t-test results

## 5 Conclusion and Recommendations

Three TO-BE scenarios were evaluated against the AS-IS baseline using paired t-tests at the 5% significance level. The combined scenario yields the greatest improvement, reducing mean cycle time by 12.11 minutes relative to AS-IS, while also producing the narrowest confidence interval, indicating a more consistent process flow. Scheduling alone reduces mean cycle time by 6.71 minutes and pre-processing by 5.47 minutes, confirming that both interventions contribute independently and that their effects are additive when combined. Based on these results, Dr. Weems is advised to pursue both interventions simultaneously. The scheduling optimization only requires administrative coordination, making this a natural first step. Pre-processing reduces load on Attending but requires organizational commitment from residents and attending physicians. As total patients processed remains the same in these scenarios, the improvements reflect reduced waiting time rather than increased throughput. This suggests that further gains would require more direct intervention at the Attending bottleneck.

## References

- Chambers, C., & Williams, K. (2017). Case – miller pain treatment center. *INFORMS Transactions on Education*, 17(3), 121–127. <https://doi.org/10.1287/ited.2017.0176cs>
- Laguna, M., & Marklund, J. (2019). *Business process, modeling, simulation and design*. CRC Press.

## A Process Flow

| Process Activity      | Resource                       | Patient type    | Data                          |
|-----------------------|--------------------------------|-----------------|-------------------------------|
| <b>Administrative</b> |                                |                 |                               |
| Sign in               | Front desk                     | All             | New: 30, Other: 15 min        |
| Register              | Front desk                     | All             | Avg: 4.33 min                 |
| Check-Out             | Front desk                     | All             | Avg: 4.72 min                 |
| <b>Clinical</b>       |                                |                 |                               |
| Vitals                | CA, Exam room                  | All             | Avg: 3.53 min                 |
| Resident review       | Resident                       | New/Return      | New: 10.44, Return: 9.30 min  |
| Resident & Patient    | Resident, Exam room            | New/Return      | New: 20.16, Return: 12.99 min |
| Teach                 | Resident, Attending            | New/Return      | New: 7.85, Return: 5.17 min   |
| Resident & Attending  | Resident, Attending, Exam room | New/Return      | New: 12.45, Return: 9.24 min  |
| <b>Follow-up</b>      |                                |                 |                               |
| PA consultation       | PA, exam room                  | Follow-up       | –                             |
| PA & Attending        | PA, Attending                  | Follow-up (50%) | Approx. 3 min                 |

Table A.1: Process activities

## B Flow chart

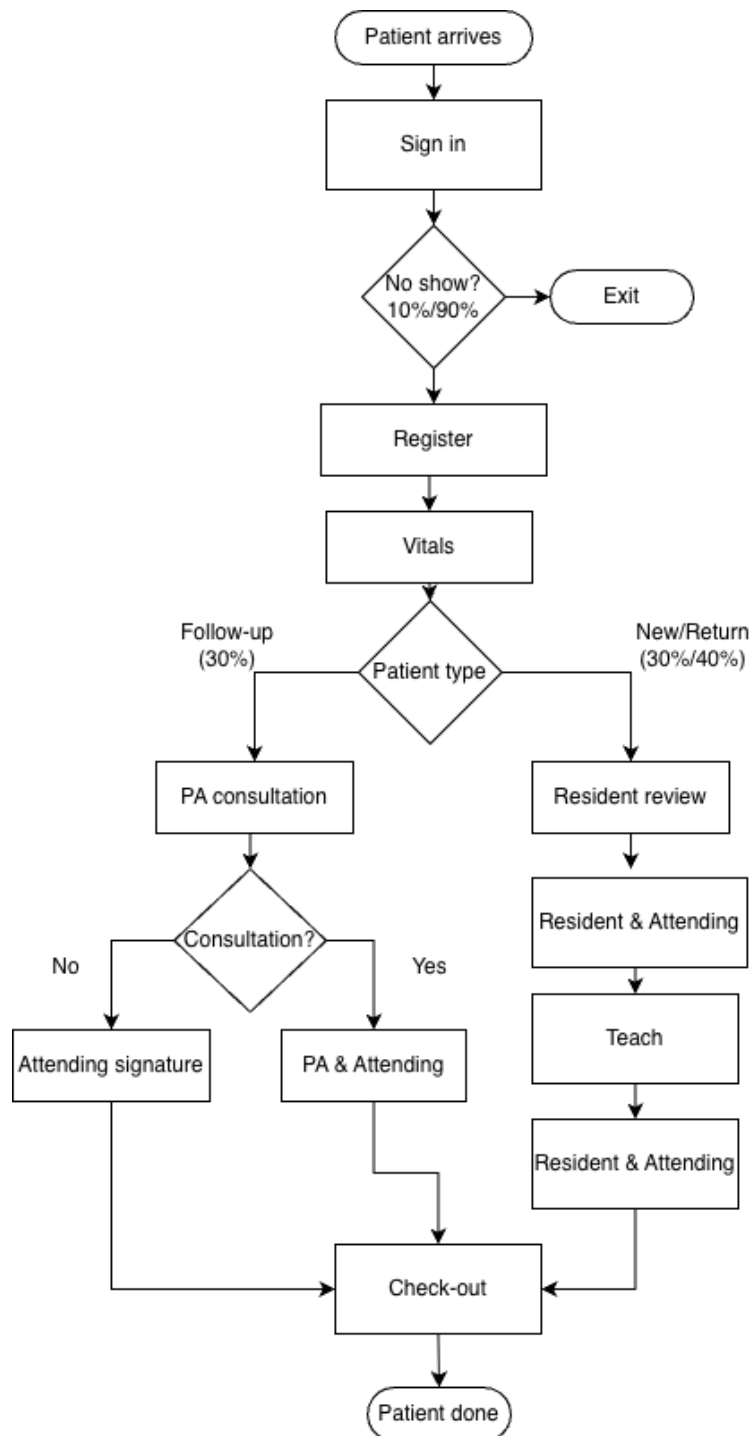


Figure B.1: Flow chart

## C Conceptual design

| Components                       | Simulation implementations     | Specifications & resources                                                                                  |
|----------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Arrival and routing</b>       |                                |                                                                                                             |
| Patient                          | SimEntity                      | One entity type. Patient type decided by PatientType Branch                                                 |
| Scheduled arrivals               | EntityGenerator, EventSchedule | 17 patients per schedule; max=17; cycle=10 h                                                                |
| No-show decision                 | Branch                         | P(continue)=0.90 → Registration; P(no-show)=0.10 → exit; sign in instantaneous                              |
| Patient type routing             | Branch                         | P(new)=0.30, P(return)=0.40, P(follow-up)=0.30                                                              |
| PA consultation?                 | Branch                         | P(consult)=0.50, P(not)=0.50                                                                                |
| <b>Common input flow</b>         |                                |                                                                                                             |
| Register                         | Server, Queue                  | Gamma(mean=4.333 min, shape=3.069)                                                                          |
| Vitals                           | Server, Queue                  | Weibull(scale=3.982 min, shape=2.107)                                                                       |
| <b>New and return flow</b>       |                                |                                                                                                             |
| Resident Review                  | Seize, Server, Queue           | Seize resident & exam room. New: Gamma(mean=10.4). Return: LogNormal( $\mu=1.737$ , $\sigma=1.002$ )        |
| Resident & Patient               | Server, Queue                  | New: LogNormal( $\mu=2.857$ , $\sigma=0.558$ ), Return: Weibull(scale=14.665, shape=1.740) → SeizeAttending |
| Teach                            | Seize, Server, Queue           | SeizeAttendingResource. New: Gamma(mean=7.861), Return: Weibull(scale=5.438)                                |
| Resident & Attending             | Server, Queue                  | New: Gamma(mean=12.450), Return: LogNormal( $\mu=1.942$ , $\sigma=0.738$ ); release all → check-out         |
| <b>Follow-up flow</b>            |                                |                                                                                                             |
| PA                               | Seize, Server, Queue           | Seize PA room. LogNormal( $\mu=2.968$ , $\sigma=0.541$ )                                                    |
| PA & Attending                   | Seize, Server, Queue           | Seize Attending; constant=3 min; release → check-out                                                        |
| <b>Departure flow and output</b> |                                |                                                                                                             |
| Check-Out                        | Server, Queue                  | Gamma(mean=4.724, shape=2.320) → PatientStatistics → Exit                                                   |
| Output                           | Statistics, EntitySink         | PatientStatistics collect total time per patient                                                            |
| <b>Resources</b>                 |                                |                                                                                                             |
| Attending                        | Resource                       | Capacity=1 (New, Return, Follow-up)                                                                         |
| Residents                        | Resource                       | Capacity=3; held from Resident Review to Resident & Attending                                               |
| Exam room (New/Return)           | Resource                       | Capacity=3; held from Resident Review to Resident & Attending                                               |
| PA exam room                     | Resource                       | Capacity=1; dedicated Follow-up; held from PA to PA & Attending                                             |

Table C.1: Simulation components, implementations, specifications and resources



## D Stochastic variables

| Variable                                   | Type       | Description                                                     |
|--------------------------------------------|------------|-----------------------------------------------------------------|
| <b>Service times</b>                       |            |                                                                 |
| Registration time                          | Continuous | Time spent on administrative registration                       |
| Vitals time                                | Continuous | Time to record patient vitals                                   |
| Physician assistant time                   | Continuous | Time spent by PA on follow-up patients                          |
| Check-out time                             | Continuous | Time to complete visit and paperwork                            |
| Resident review (new)                      | Continuous | Time reviewing patient file (new patients)                      |
| Resident–patient<br>interaction (new)      | Continuous | Consultation time with resident                                 |
| Teaching time (new)                        | Continuous | Teaching interaction between resident and attending             |
| Resident–attending<br>interaction (new)    | Continuous | Discussion between resident and attending                       |
| Resident review (return)                   | Continuous | Time reviewing file (return patients)                           |
| Resident–patient<br>interaction (return)   | Continuous | Consultation time with resident                                 |
| Teaching time (return)                     | Continuous | Teaching interaction for return patients                        |
| Resident–attending<br>interaction (return) | Continuous | Discussion between resident and attending                       |
| <b>Routing decisions</b>                   |            |                                                                 |
| Patient type                               | Discrete   | New, return, or follow-up patients                              |
| Follow-up      requires<br>attending       | Discrete   | Whether PA consults attending (approximately 50%)               |
| No-show / cancellation                     | Discrete   | Whether patient attends appointment (approximately 10% no-show) |

Table D.1: Identified stochastic inputs

## **E JaamSim documentation**

### **E.1 AS-IS**

The AS-IS model simulates the Miller Pain Treatment Center in its current state. Patients arrive at their scheduled appointments, with 17 appointments between 08:00 and 11:15. A 10% no-show rate is applied with a branch at arrival.

After registration and vitals, patients are routed by type. New and return patients seize a resident and an examination room before going through resident processes with attending and teaching. Follow-up patients seize an examination room and are seen by a physician assistant, with a 50% probability of needing additional medical attention.

Service time distributions were fitted from the data provided in the case given in Chambers and Williams, 2017, with the distributions given in Table 3.1.

### **E.2 TO-BE 1: Scheduling**

The TO-BE 1 (Scheduling) is identical to the AS-IS model in its structure. The only modification done is changing of the patient schedule, where all appointments are evenly spread every 15 minutes throughout each session. This is done after Dr. Weems' suspicion that clustered appointments contribute to inefficiency in the system (Chambers & Williams, 2017).

### **E.3 TO-BE 2: Pre-processing**

Dr. Weems also suspects that the variance across teaching contributes to performance loss in the system, suggesting that reviewing the case the day before might decrease waiting times in this area. Dr. Weems estimates that this could cut average review- and teaching times in half (Chambers & Williams, 2017). We have therefore reduced distributions connected to these processes, while all other variables are kept intact.

### **E.4 TO-BE 3: Combined**

The combined model simulates a system where both scheduling and pre-processing are used, to see if a combination of the two modifications provides better performance than either alone.